Seat No: .... **Enrollment No:** ....

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech/Int. B.Tech, Summer 2025-26 Examination

Semester: 5/9 **Date: 09-05-2026**
Subject Code: 303105306/203105305 **Time: 10:30 AM To 01:00 PM**
Subject Name: Theory of Computation/Formal Language & Automata Theory **Total Marks: 60**

Instructions:
1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1** is compulsory, From Section B, **Q.1** is compulsory.
3. Figures to the right indicate full marks
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

SECTION - A

Q.1	Answer the following questions.	**Marks**	**CO**	**BT**
	A. a) Explain string homomorphisms with an example. b) Find h(L) if h(a)=01, h(b)=10, and L={a,b}. c) Prove that the homomorphic image of a regular language is also regular.	06	CO1	BT1
	B. a) Explain the Chomsky hierarchy of grammars. b) Give examples for each type of grammar. c) Show inclusion relations among the four language classes.	06	CO1	BT2
Q.2	A. Construct a regular expression for all strings over Σ={0,1} containing exactly two 1s.	04	CO2	BT4
	B. Explain the practical applications of Mealy and Moore machines in digital circuit design.	05	CO2	BT1
	OR			
	B. Convert the ε-NFA with δ(q0,ε)={q1,q2}, δ(q1,a)={q1}, δ(q2,b)={q2} into an equivalent DFA.	05	CO2	BT1
Q.3	A. Define a regular language and give two examples.	04	CO2	BT2
	B. Show that CFGs can generate ambiguous languages with an example.	05	CO3	BT4
	OR			
	B. Construct a PDA for L={ a^n b^n | n≥0}.	05	CO3	BT1

SECTION - B

Q.1	Answer the following questions.	**Marks**	**CO**	**BT**
	A. 1.Give an example of a context-sensitive production rule. 2.What is the role of context in CSG rules? 3.Is every CFG also a CSG? Explain briefly.	06	CO3	BT2
	B. 1.Give an example of a language generated by a CSG but not by a CFG. 2.Define a context-sensitive language (CSL). 3.State the relation between CSGs and CSLs.	06	CO3	BT1
Q.2	A. Write the components of a Turing Machine and their roles. Explain the role of accept and reject states in TM.	04	CO4	BT2

	B. Show that the complement of a Turing-recognizable language need not be Turing-recognizable.	05	CO4	BT3
	OR			
	B. Explain difference between time complexity of multitape vs single tape TM.	05	CO4	BT5
Q.3	A. Differentiate between unrestricted grammars and recursive languages.	04	CO4	BT6
	B. State and explain the Church-Turing thesis. Why is it significant in computation theory?	05	CO5	BT3
	OR			
	B. Write a note on the relationship between recursive languages and recursively enumerable languages in the context of undecidability.	05	CO5	BT3